

Lab Experiment:

Virtualization in Cloud Computing

Learning Objectives

By the end of this lab, students will be able to:

1. Explain the difference between Type 1 and Type 2 hypervisors and where each is used in cloud data centres.
2. Create and configure a virtual machine (VM) using a Type 2 hypervisor (VirtualBox).
3. Allocate and modify virtual resources (CPU, RAM, disk, network) and observe their effect.
4. Take and restore VM snapshots, and clone a VM.
5. Relate desktop virtualization concepts to how public cloud providers (e.g., AWS EC2) provision virtual machines at scale.
6. Compare virtualization to containerization at a conceptual level.

Materials Needed

- A laptop/desktop with virtualization support enabled in BIOS/UEFI (Intel VT-x or AMD-V), 4+ GB free RAM, 15+ GB free disk space
- VirtualBox (free) installed, or an equivalent hypervisor already installed in your lab image
- A lightweight OS ISO (e.g., Ubuntu Server, Ubuntu Desktop minimal, or Lubuntu — any small Linux ISO works well for time)
- (Optional, Part 4) A free-tier AWS, Azure, or GCP account — instructor should confirm in advance whether this is available for the session

Part 1 — Concepts: Hypervisors and Virtualization (10 minutes)

Read the following, then answer the questions before moving on.

Virtualization is the creation of a virtual (rather than physical) version of a computing resource — a server, storage device, network, or OS. The software layer that makes this possible is the hypervisor, also called a Virtual Machine Monitor (VMM).

	Type 1 (Bare-Metal)	Type 2 (Hosted)
Runs on	Directly on physical hardware	On top of a host operating system
Examples	VMware ESXi, Microsoft Hyper-V, Xen, KVM	VirtualBox, VMware Workstation, Parallels
Performance	Higher (less overhead)	Lower (extra OS layer)
Typical use	Data centers, cloud providers (AWS, Azure, GCP)	Developer laptops, testing, labs

Check-in questions (write brief answers in your lab notebook/report):

1. Which type of hypervisor do you think AWS, Azure, and GCP run on their physical servers, and why does that matter for performance and multi-tenancy?
2. What is the difference between a hypervisor and an operating system?
3. Name one advantage and one disadvantage of virtualization for a cloud provider.

Part 2 — Hands-On: Creating Your First Virtual Machine (25 minutes)

Step 1: Verify virtualization support (2 min)

- Windows: Open Task Manager → Performance tab → CPU. Confirm "Virtualization: Enabled."
- Mac/Linux: Run `sysctl -a | grep machdep.cpu.features` (Mac, look for VMX) or `egrep -c '(vmx|svm)' /proc/cpuinfo` (Linux, result > 0 means supported).

Step 2: Install VirtualBox (5 min, skip if pre-installed)

Download and install VirtualBox from [virtualbox.org](https://www.virtualbox.org) for your OS. Accept default settings during installation.

Step 3: Create a new Virtual Machine (8 min)

1. Open VirtualBox → click New.
2. Name the VM (e.g., CloudLab-VM1), select Type: Linux, Version: matching your ISO.
3. Allocate RAM: 1024–2048 MB (leave enough for your host OS).
4. Create a virtual hard disk: choose VDI, dynamically allocated, 10–15 GB.
5. In Settings → Storage, attach your downloaded ISO to the virtual optical drive.
6. In Settings → System → Processor, allocate 1–2 virtual CPUs.
7. In Settings → Network, confirm Adapter 1 is set to NAT (default) — note this is how the VM gets internet access through the host.

Step 4: Boot and install the OS (8 min)

1. Click Start. The VM boots from the ISO.
2. Proceed through the OS installer with default options (you do not need to complete a full install if time is short — booting into the live/installer environment is sufficient to demonstrate the concept).
3. Once at a desktop or terminal prompt, run a basic command (e.g., `top`, `lscpu`, or `free -h` on Linux) to view the resources as seen from inside the VM.

Step 5: Compare host vs. guest resources (2 min)

Fill in this table in your report:

Resource	Host (your real machine)	Guest (VM)
CPU cores visible		
RAM visible		
Disk size visible		

Reflection: Why do the numbers differ, and who is responsible for enforcing that the guest cannot see or use more than what was allocated?

Part 3 — Snapshots and Cloning (12 minutes)

Snapshots and cloning are core to how cloud providers offer fast provisioning, rollback, and scaling.

Step 1: Take a snapshot (4 min)

1. With the VM running (or powered off), go to Machine → Take Snapshot.
2. Name it baseline-state.
3. Make a visible change inside the VM (e.g., create a file called test.txt on the desktop, or change the desktop background).

Step 2: Restore the snapshot (4 min)

1. Power off the VM.
2. Go to the Snapshots tab, right-click baseline-state, choose Restore.
3. Boot the VM again and confirm your change (e.g., test.txt) is gone.

Reflection: How is this similar to how a cloud provider might let you "roll back" a server, or spin up a new instance from a saved machine image (e.g., an AWS AMI)?

Step 3: Clone the VM (4 min)

1. Right-click your VM → Clone.
2. Choose "Full Clone" and create a new VM named CloudLab-VM2.
3. Briefly start the clone to confirm it boots independently.

Reflection: In a public cloud, this cloning operation happens automatically when you increase the number of instances in an auto-scaling group. What manual steps did you just do that a cloud platform automates?

Part 4 — Connecting to the Cloud (15 minutes)

Choose Option A if your instructor has provided cloud credentials/free-tier access; otherwise complete Option B.

Option A: Launch a cloud VM (hands-on)

1. Log into the AWS Management Console (or Azure/GCP if specified).
2. Navigate to EC2 → Launch Instance.
3. Choose a free-tier eligible AMI (e.g., Amazon Linux 2023 or Ubuntu), instance type t2.micro or t3.micro.
4. Configure 1 vCPU, default storage (8 GB), and a security group allowing SSH (port 22).
5. Launch the instance and note the instance ID, public IP, and AMI ID.
6. Compare this launch process step-by-step against Part 2's VirtualBox VM creation.

Step	VirtualBox (local)	AWS EC2 (cloud)
Choosing the "image"	ISO file	AMI
Choosing CPU/RAM	Settings → System	Instance type (e.g., t2.micro)
Choosing storage	Virtual disk (VDI)	EBS volume
Networking	NAT adapter	VPC / Security Group
Hypervisor	VirtualBox (Type 2)	Nitro/Xen (Type 1)

Important: Terminate the instance at the end of the lab to avoid unwanted charges.

Option B: Case study walkthrough (no cloud account needed)

Read the following scenario and answer the questions:

A company runs its e-commerce website on physical servers. Each server is only used at 15% average CPU utilization, but they own 20 servers to handle rare peak-holiday traffic. They are considering moving to a cloud provider that uses virtualization to run many customers' VMs on shared physical hardware.

Questions:

1. How does virtualization let the cloud provider improve hardware utilization compared to the company's current setup?
2. What is "elasticity," and how does the ability to clone/snapshot VMs (Part 3) support it?
3. What risk does multi-tenancy (many customers' VMs sharing one physical server) introduce, and how does the hypervisor help mitigate it?